

# CS3 Lab 3 Code Review Quiz

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Choose the best answer for each question. Assume the Lab 3 `node<T>` and `Collection<T>` implementations are used exactly as provided.

## 1. What does `Collection<int> numbers;` demonstrate?

- A. A collection that can store values of any type at runtime
- B. A template instantiation whose item type is `int`
- C. A node that can store only one integer permanently
- D. A collection that cannot be copied

## 2. After the following statements execute, what does `items.last()` return?

cpp

```
Collection<int> items;
```

```
items.add(4);
```

```
items.add(7);
```

```
items.add(2);
```

- A. 4
- B. 7
- C. 2
- D. It throws `std::out_of_range`

## 3. What happens after this call?

```
```cpp
```

```
Collection<int> values;
```

```
values.add(3);
```

```
values.add(1);
```

```
values.add(3);
```

```
values.remove(3);
```

- A. The collection contains 3 1
- B. The collection contains 1 3
- C. The collection contains only 1
- D. The program throws because duplicate values are not allowed

## 4. What does `last()` do when called on an empty collection?

- A. Returns `nullptr`
- B. Returns a default-constructed `T`

- C. Prints an error and returns -1
- D. Throws `std::out_of_range`

## 5. What is printed by the following code?

cpp

```
Collection<std::string> words;  
words.add("red");  
words.add("blue");  
words.print();
```

- A. ``redblue`` with no spaces or newline
- B. ``red blue`` followed by a newline
- C. ``blue red`` followed by a newline
- D. The memory addresses of the two nodes

## 6. Why does the copy constructor add each item from ``other`` instead of simply assigning ``head_``?

- A. To preserve the order while creating separate nodes
- B. To make the copied collection automatically sorted
- C. To prevent ``last()`` from working on the copy
- D. To convert every item to a string

## 7. Given the following code, what is the purpose of ``std::swap(head_, copy.head_)`` in the assignment operator?

```cpp

```
Collection<int> first;  
Collection<int> second;  
first.add(10);  
second = first;
```

- A. It swaps the values of the first and last nodes in first
- B. It gives second the copied list and lets the temporary clean up the old list
- C. It deletes the source collection before copying it
- D. It makes first and second share the same nodes

## 8. Why is this statement safe in the provided assignment operator?

cpp

```
collection = collection;
```